LINEAR QUEUE USING ARRAY - C PROGRAM

1. PROGRAM CODE

```c
#include <stdio.h>
#define SIZE 5

int queue[SIZE];
int front = -1, rear = -1;

void enqueue()
{
    int x;
    if (rear == SIZE - 1)
    {
        printf("Queue Overflow\n");
        return;
    }

    printf("Enter element: ");
    scanf("%d", &x);

    if (front == -1)
        front = 0;

    queue[++rear] = x;
    printf("%d inserted into queue\n", x);
}

void dequeue()
{
    if (front == -1)
    {
        printf("Queue Underflow\n");
        return;
    }

    printf("%d deleted from queue\n", queue[front++]);

    if (front > rear)
        front = rear = -1;
}

void display()
{
    int i;
    if (front == -1)
    {
        printf("Queue is empty\n");
        return;
    }

    printf("Queue: ");
    for (i = front; i <= rear; i++)
        printf("%d ", queue[i]);
    printf("\n");
}

int main()
{
    int choice;

    while (1)
    {
        printf("\n1. Enqueue\n2. Dequeue\n3. Display\n4. Exit\nEnter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1: enqueue(); break;
            case 2: dequeue(); break;
            case 3: display(); break;
            case 4: printf("Exiting...\n"); return 0;
            default: printf("Invalid choice\n");
        }
    }
}
```

2. ACTUAL PROGRAM OUTPUT

Representative full run (captured stdout)

```
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: Enter element: 10 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: Enter element: 20 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: Queue: 10 20

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 10 deleted from queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: Queue: 20

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: Exiting...
```